McDonald's Financial Forecast Project

Professional Summary & Technical Report

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Excel Version: March 2025 Python Version: October 2025

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Abstract

This project analyses McDonald's historical performance (2014–2018) and generates 5-year forward projections (2019–2023) under three scenarios: Baseline Growth, Pandemic Disruption, and Hindsight. The model was built first as a comprehensive Excel financial model (March 2025) and later fully replicated in Python with interactive Plotly visualisations (October 2025). Using pre-COVID data only, the exercise served as a forward-looking case study of a severe global crisis. All monetary values are CPI-adjusted to 2018 USD (later validated to 2025 USD in outcome analysis). Post-hoc validation against actual 2019–2023 results shows the model captured the initial disruption directionally well while modestly underestimating McDonald's resilience through delivery and digital channels.

1 Executive Summary

The McDonald's Financial Forecast Project demonstrates rigorous scenario-based financial modelling and post-event validation. Historical data (2014–2018) from 10-K and 10-Q filings were used to project 2019–2023 performance under three scenarios:

- **Baseline Growth**: Steady 4 % market CAGR and 2 % annual revenue growth.
- **Pandemic Disruption**: Severe shock modelled on 2003 SARS and 2008–2009 crisis patterns (market CAGR 1.5 %, explicit revenue declines of -10 % in 2020 and -14 % in 2021).
- **Hindsight**: Optimistic relative to the Pandemic scenario (market CAGR 2 %) but still realistic, sharing the same revenue shock assumptions to isolate company-specific vulnerability.

Outcome analysis (Part 2) compared projections with actual 2019–2023 data (all adjusted to 2025 USD). Average errors were modest: +5.81 % (Pandemic market), +7.41 % (Hindsight market), and -0.41 % (revenues in both disruption scenarios). The project highlights the value of dynamic scenario modelling for stress-testing resilience in the quick-service restaurant sector.

2 Project Overview & Data

- **Data Period**: Historical 2014–2018 (used for model calibration); actual outcomes 2019–2023 (used for validation).
- **Sources**: McDonald's 10-K and 10-Q filings (SEC EDGAR), IBISWorld Global Fast Food Market data, BLS CPI-U.

- **Adjustments**: All monetary figures CPI-adjusted to 2018 USD; common stock left unadjusted (share count, not monetary value). Later validated to 2025 USD in outcome analysis.
- **Implementation**: Excel version completed March 2025; fully interactive Python version (pandas + Plotly) completed October 2025.

The model replicates a complete Excel financial dashboard with transposed tables, ratio analysis, segment breakdowns, quarterly seasonality/volatility, market research, assumed growth assumptions, forecasts, and outcome validation.

3 Detailed Methodology

The project follows a structured, layered approach. Each component is described below with its **business impact** and **why the method is suitable**.

3.1 1. Historical Financial Statements (Income, Balance Sheet, Cash Flow)

Transposed tables present metrics in rows and years in columns, with CPI adjustment, CAGR calculations, and ratio derivations.

Business impact: Provides a clean, comparable view of profitability, liquidity, and cash generation over time.

Why suitable: Transposed format mirrors professional financial modelling practice; CPI adjustment ensures real (inflation-adjusted) trend analysis.

3.2 2. Ratio Analysis

Comprehensive ratios across profitability, leverage, cash-flow performance, liquidity, quality of earnings, return ratios, dividend sustainability, and market performance.

Business impact: Identifies operational efficiency, financial health, and cash-flow quality.

Why suitable: Ratios are the standard language of financial analysis; interactive Plotly visualisations allow quick trend detection.

3.3 3. Segment & Quarterly Analysis

Geographic segment breakdown (U.S., International Lead Markets, High Growth Markets, Foundational Markets & Corporate) and quarterly revenue with seasonality factors and volatility metrics.

Business impact: Reveals which regions drive growth and exposes intra-year patterns and volatility.

Why suitable: Segment data is essential for geographic risk assessment; seasonality and volatility metrics are critical for operational planning.

3.4 4. Market Research & Assumed Growth

Global fast-food market sizing, McDonald's market share, and explicit growth assumptions for each scenario.

Business impact: Grounds forecasts in external market context and stress-tests resilience under different economic regimes.

Why suitable: Scenario planning is industry best practice for forward-looking risk management; assumptions are explicitly derived from historical crises (SARS, 2008–2009).

3.5 5. Forecasting

5-year projections (2019–2023) of market size, revenues, and market share under all three scenarios.

Business impact: Translates assumptions into quantifiable financial outcomes.
Why suitable: Deterministic scenario modelling combined with CPI adjustment produces transparent, auditable forecasts.

3.6 6. Outcome Analysis & Validation

Direct comparison of projected vs. actual 2019–2023 data (adjusted to 2025 USD) with percentage error calculations.

Business impact: Quantifies model accuracy and highlights where McDonald's outperformed expectations (delivery/digital resilience).
Why suitable: Post-hoc validation is the gold standard for assessing forecasting quality; average errors provide clear performance metrics.

4 Outcome Analysis Results

All figures adjusted to 2025 USD for consistency.

Metric	Pandemic Disruption	H
Average Error (2019–2023)	+5.81%	
Interpretation	Modest underestimate of market resilience	Modest underestimate of market r

Table 1: Outcome Analysis – Average Percentage Errors vs. Actual 2019–2023 Data

The model captured the initial disruption directionally well but modestly underestimated the speed of McDonald's recovery, largely due to unmodelled strength in delivery and digital channels.

5 Parameter Calibration & References

All assumptions were grounded in:

- McDonald's official 10-K and 10-Q filings (2014–2023).
- IBISWorld Global Fast Food Restaurants market data.
- BLS CPI-U historical indices and 1 % annual inflation assumption for 2019–2023 multipliers.
- Historical crisis benchmarks (2003 SARS, 2008–2009 financial crisis).

Full references (with direct SEC links) are included in the project files.

6 Business & Strategic Impact

The project demonstrates how pre-crisis data can be used to stress-test a major QSR operator against a severe global health disruption. Key takeaways:

- Scenario planning effectively quantified downside risk and recovery potential.
- McDonald's showed greater resilience than the broader market during the modelled pandemic period.

- The framework is directly transferable to other retailers or QSR chains for capital allocation and risk management.

Completed as a self-taught case study in 2025, the work showcases advanced financial modelling, scenario analysis, and rigorous post-hoc validation.

7 Conclusion

The McDonald's Financial Forecast Project, spanning 2014-2023, analyzed historical financial data and projected revenues under three scenarios - Baseline, Pandemic Disruption, and Hindsight - adjusted to 2025 USD for relevance. Part 2 (Outcome Analysis) validated these projections against actual 2019-2023 data, revealing average errors of 5.81% for the Pandemic Disruption global market, 7.41% for the Hindsight global market, and -0.41% for revenues in both scenarios, indicating a slight underestimate likely due to McDonald's unmodeled resilience through delivery trends. This project highlights the importance of dynamic modeling in forecasting, paving the way for future improvements through advanced techniques.